

SENONI RESEARCH

Behavioral Compilation

From Harness to Weights on a Single Machine

A Governed Study of Narrow LLM Specialization on Apple Silicon,
Inspired by Production-Scale Agent Learning

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Research status. This is a systems research working paper, not a peer-reviewed publication and not a deployment claim. It reports a bounded local experiment, including failed runs, post-hoc diagnostics, matched controls, semantic defects, and unresolved validation limits. Private case evidence and model weights are not released.

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Abstract

Production AI systems often accumulate domain expertise in discrete artifacts surrounding a frontier model: system prompts, tool descriptions, retrieval rules, schemas, validators, routing logic, and application code. Recent Shopify engineering reports describe a transition from such a harness toward continual parameter updates that compress recurring production lessons into specialized model weights. This paper asks a smaller and more falsifiable question: *can the methodological structure of that transition be studied meaningfully on a single local workstation, without confusing formatting gains, memorization, or evaluator defects with learned task behavior?*

We report a sequence of governed experiments on one Apple M4 Max with 128 GB unified memory using MLX/MLX-LM and a four-bit Qwen3-4B model. An initial closed-book acquisition smoke test was dominated by full-context and oracle controls, motivating a shift from “company facts in weights” toward narrow evidence-assisted task specialization. We then trained a rank-16 QLoRA adapter on five approved long-context deliverable-review examples, eight exposures each, for exactly 40 optimizer updates. The longest complete row contained 35,406 tokens. A stock long-context backward pass exhausted Metal memory; a training-only runtime using query-blocked causal attention and supervised-position output projection completed the workload, subject to explicitly documented numerical-equivalence and memory-accounting qualifications.

The first unconstrained evaluation formally failed: 0/5 complete valid outputs because every answer began with unexpected protocol markers. A separate body-only diagnostic nevertheless found 25/25 requirement labels and 5/5 dispositions. To isolate interface effects from weight effects, we then compared the unadapted base and the unchanged adapter under the same schema-constrained decoder. Both produced 5/5 structurally valid reviews; the base achieved 22/25 labels and 3/5 dispositions, while the adapter achieved 25/25 and 5/5. A subsequent blinded model-advisory review found unsupported source claims, authority errors, and one contradictory next action in the ten-answer pool; incomplete frozen evidence prevented overall semantic pass/fail scores.

The central result is therefore deliberately narrower than “enterprise memory.” The adapter learned useful decision boundaries on the examples it saw, while reliable evidence-grounded review remained unestablished. From this result we propose *behavioral compilation*: stable organizational behavior may be compiled into weights, while mutable authority remains external and addressable. We formalize this as a three-plane architecture—evidence, behavioral weights, and control—and a layered *Specialization Validity Stack* that separates provenance, structural validity, decision correctness, evidence support, authority safety, generalization, and continual stability. The single-machine setting is not presented as a substitute for production scale. It is presented as a scientific microscope: small enough to freeze, hash, rerun, falsify, and attribute improvements to the harness or the weights rather than to both at once.

Keywords: large language models; local fine-tuning; QLoRA; MLX; Apple Silicon; task specialization; continual learning; structured generation; evidence grounding; enterprise AI; behavioral compilation.

Central thesis. In governed enterprise AI, model weights should not be treated as the authoritative database. A more defensible role is to compile *stable behavioral priors* into parameters while keeping mutable facts, permissions, version history, and audit evidence in external systems that remain directly addressable.

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1 Introduction

The dominant enterprise pattern for large language models is still open-book: retrieve documents, inject context, call tools, and constrain what the model may do. This pattern is rational. Retrieval-augmented generation keeps source material mutable and inspectable [Lewis et al., 2020]; tool and policy layers retain operational control; and application code can enforce constraints that a probabilistic model should not own.

At the same time, production systems accumulate an increasingly large behavioral harness around the model. Prompts describe policy. Tool schemas encode domain vocabulary. Routers embody task decomposition. Validators encode failure conditions. Examples define acceptable responses. Each correction adds another discrete artifact. Shopify describes this explicitly: early improvements accumulate in prompts, context, tool definitions, and orchestration, while the deployed model remains frozen; its continual-learning work then attempts to translate recurring production lessons into parameter updates [McNamara and Mazza-Anthony, 2026]. Shopify’s Flow work similarly reports moving a production skill from a frontier model toward a fine-tuned Qwen3-32B agent, with representation design, training-serving parity, calibrated evaluation, and a production feedback loop proving as important as the optimizer itself [Chaiwachirasak and colleagues, 2026].

That industrial trajectory suggests a research question that is broader than fine-tuning accuracy and narrower than artificial general intelligence:

Which parts of an organization’s repeated behavior should remain explicit in the harness, and which parts can be safely compressed into model parameters?

The question is easy to ask at production scale and difficult to study causally. When a system changes prompts, tools, data, model weights, decoder constraints, evaluator prompts, and traffic distribution at once, a higher score may be real but attribution is weak. A smaller environment can sometimes provide stronger experimental discipline. In this work, one workstation becomes a laboratory for separating three sources of improvement:

1. **Harness improvement:** the surrounding system makes the same model easier to steer or validate.
2. **Weight improvement:** the model’s parameters change its task decisions under an otherwise matched interface.
3. **Evaluation improvement:** the measurement becomes more faithful and stops rewarding or penalizing the wrong behavior.

The paper reports a sequence of experiments in which all three occurred, sometimes in opposite directions. An early parametric acquisition candidate improved over the base model but was decisively outperformed by full-context controls. A later five-example specialization run drove training loss low and produced target-matching decision labels, yet failed its original output contract because of unexpected protocol tokens. Schema-constrained generation fixed the interface for both the base and adapted model. Under that identical interface, the adapter recovered three labels and two case dispositions that the base missed. A later content review showed that correct labels still coexisted with unsupported claims and unsafe or contradictory operational language.

This sequence motivates two proposed contributions beyond the specific experiment.

First, we introduce **behavioral compilation** as a design framing. The source of truth remains external. Training examples, rubrics, and approved trajectories act like source material. Parameter-efficient post-training compiles recurring task behavior into a model artifact. Structured decoding and deterministic checks act as a type and interface boundary. Evaluation remains responsible for determining whether the compiled behavior preserves the semantics that matter.

Second, we introduce a **Specialization Validity Stack**: a result is not a scalar until one deliberately chooses what to collapse. Provenance integrity, valid structure, correct decisions, supported evidence, authority-safe actions, counterfactual sensitivity, and continual retention are separate dimensions. Our

own experiment reached different states on different layers, which is precisely why a single headline accuracy would have been misleading.

The empirical claims are intentionally modest. We did not evaluate generalization to unseen real work. We did not demonstrate closed-book company knowledge acquisition. We did not certify semantic safety. We did not prove the custom long-context runtime numerically identical to the stock path at production lengths. We did show that a bounded, long-context, exact-example QLoRA experiment can be executed locally; that a matched constrained decoder can separate interface gains from adapter gains; and that semantic inspection can invalidate the intuitive leap from “all labels correct” to “the review is correct.”

2 Industrial Motivation: What We Borrowed from Shopify

Shopify’s engineering publications are industrial reports rather than peer-reviewed papers, but they are unusually useful for systems research because they expose failure modes that conventional benchmark papers often omit: train-serving drift, representation mismatch, insufficient refusal data, judge calibration, reward hacking, and production/offline gaps.

2.1 From a harness to parameter updates

Shopify’s continual-learning account draws a useful boundary between *discrete artifacts* and *continuous parameter updates*. The discrete side includes prompts, context, tool definitions, and control flow; the parameter side begins only after the harness has been pushed far enough and hard negatives are available [McNamara and Mazza-Anthony, 2026]. This sequencing influenced our decision not to train immediately. We first spent several iterations correcting task representation and evaluation semantics, including cases where deterministic substring logic incorrectly judged natural-language obligations.

The same report places the quality rubric before the training loop. Product requirements are decomposed into dimensions such as completeness, execution quality, response quality, and safety, and automated judges are calibrated against human annotations rather than trusted by default [McNamara and Mazza-Anthony, 2026]. Our local work could not reproduce that production annotation scale, but it adopted the principle that evaluator quality is part of the experiment, not a reporting afterthought.

2.2 Representation is part of the learning problem

Shopify Flow offers a particularly relevant result. The production representation was a custom JSON DSL, but the model was trained to generate a semantically equivalent Python representation that could be round-trip transpiled. With the same training data, Shopify reports materially higher syntactic and semantic correctness after moving the target representation closer to the model’s pretraining distribution [Chaiwachirasak and colleagues, 2026]. This aligns with broader work on intermediate representations for workflow and low-resource language generation [Fan et al., 2024, Mora et al., 2024, Bhatia et al., 2024].

We did not implement Shopify’s Python transpiler. Our analogous lesson was more limited: output structure and task learning must be experimentally separable. The first after-training generation failed the whole-text JSON contract even though a complete target-matching review object was present after unexpected protocol markers. A later grammar-constrained decoder removed the formatting degree of freedom for *both* base and adapted models. That enabled a matched comparison in which structural validity was held constant and only decision behavior differed.

2.3 Training-serving parity is not a cosmetic concern

Shopify reports that tool names, tool order, response field order, system prompts, and tool descriptions affected model accuracy when training and production differed [Chaiwachirasak and colleagues, 2026]. Our own audit encountered a related phenomenon: the completed-assistant chat template and the generation-prompt template were not symmetric, and the supervised suffix contained template-added thinking tokens that were absent from the generation prompt. The model then emitted a different protocol marker not present in the training messages. We do not claim the same mechanism, but both experiences support

the same systems principle: the token-level contract between training and serving must be inspected, not inferred from semantic equivalence.

2.4 Negative cases are first-class behavior

Shopify’s refusal-curation work identifies a structural bias in production data: successful interactions are logged and retained, while examples that should be refused may be absent. Their segmentation skill therefore had tens of thousands of successful examples but zero refusal examples, and the model improvised on impossible requests until a curated negative-data process was added [Xie, 2026]. Our five-example experiment is much smaller, but its most informative cases were likewise the rare negative states: two requirements were `insufficient_evidence`, one was `not_met`, and two cases required non-proceed dispositions. The base model already handled the majority `met/proceed` states. The adapter’s entire measured label/disposition gain came from the rare boundary cases.

2.5 The harness still outlasts the model

Shopify’s application-security work makes an apparently opposite point: a strong harness can matter more than model choice. Test oracles, cross-model verification, deterministic scripts, and bounded context partitions persist while models change [Deveau, 2026]. This is not in tension with continual learning. It suggests a division of labor: some behavior is worth compiling into weights, but external determinism should continue to own credentials, storage, schemas, provenance, and hard operational constraints.

Table 1: Methodological downscaling: production-scale lessons versus our local laboratory. This is not a reproduction of Shopify’s performance or production system.

Dimension	Shopify examples	Local study in this paper
Model scale	Qwen3-32B Flow agent; other specialized open models in production	Qwen3-4B Instruct, four-bit, rank-16 QLoRA
Compute	Flow training reported on two H200 nodes with a 12-hour weekly run	One M4 Max, 128 GB unified memory; exactly 40 measured optimizer updates
Data	Thousands to tens of thousands of production or production-derived examples; recurring flywheel	Five approved development-reference examples, each exposed eight times
Representation	Python intermediate representation transpiled to production JSON DSL	Native review JSON, later enforced through schema-constrained generation
Evaluation	Programmatic checks, calibrated LLM judges, human labels, production traffic	Strict schema and decision scoring, matched base control, blinded model-advisory semantic review
Hard negatives	Refusals, impossible requests, ambiguous cases mined and curated	Two <code>insufficient_evidence</code> , one <code>not_met</code> , two non-proceed dispositions
Continual loop	Production failures become repaired trajectories, SFT/RL data, and future training	Not reproduced; explicitly deferred until a trustworthy rubric and held-out evidence exist
Deployment	Production agents serving live merchant traffic	No deployment, no promotion, no generalization claim

The intent was therefore not to reproduce Shopify at laptop scale. It was to preserve the *experimental anatomy*: isolate the skill, define quality, control the representation, preserve hard negatives, compare against a matched baseline, and inspect what the weights contribute after the harness has been held fixed.

3 Research Context

3.1 Parameter-efficient specialization

LoRA freezes pretrained weights and learns low-rank updates, making downstream adaptation substantially cheaper than full fine-tuning [Hu et al., 2021]. QLoRA adds quantized frozen base weights and backpropagates into low-rank adapters, demonstrating that substantial models can be adapted under much smaller memory budgets [Dettmers et al., 2023]. These methods make local experimentation plausible, but they do not imply that LoRA and full fine-tuning learn equivalent internal solutions. Later analysis finds distinct spectral structure and different out-of-distribution behavior, especially in sequential adaptation [Shuttleworth et al., 2024]. We therefore treat QLoRA as an experimental mechanism, not as a transparent substitute for full fine-tuning.

Small amounts of high-quality supervision can strongly shape model behavior. LIMA showed that carefully selected instruction examples can induce substantial alignment behavior with relatively little supervised data [Zhou et al., 2023]. Distilling Step-by-Step showed that rationales from stronger models can provide efficient supervision to smaller models [Hsieh et al., 2023]. These results motivated our interest in a tiny, curated development-reference set—but neither paper licenses a generalization claim from five examples.

3.2 Knowledge in context versus knowledge in weights

RAG provides a well-established pattern for combining parametric generation with external, non-parametric memory [Lewis et al., 2020]. The attraction of parametric knowledge is different: retrieval-free latency, reduced prompt length, and the possibility of specialized internal behavior. Model-editing work such as MEND, ROME, and MEMIT demonstrates that targeted factual changes in parameters are feasible [Mitchell et al., 2021, Meng et al., 2022a,b]. The harder question is not whether a weight can change, but whether the resulting knowledge remains reachable, composable, governable, and stable under subsequent changes.

Three 2026 preprints were particularly influential in refining our hypothesis. IAR separates document *injection*, QA *alignment*, and general-capability *recovery*, arguing that retrieval-free knowledge internalization is itself multi-stage [Kou et al., 2026]. O’Neill finds that diverse study representations create more usable factual knowledge than bare statements, yet sequential later writes can make earlier facts behaviorally inaccessible; supplying the fact in context remains a substantially more reliable channel [O’Neill, 2026]. Pulici et al. show that LoRA adapters can store entity-local knowledge while identifying adapter selection as a separate retrieval problem: storage does not solve addressing [Pulici et al., 2026].

These findings moved our project away from a simplistic proposition—“put the company corpus into weights”—toward a more conservative split. Mutable facts are better suited to external, versioned evidence. Stable task behavior may be a more defensible target for parameters.

3.3 Long-context adaptation and local hardware

Long-context fine-tuning is computationally expensive because standard attention scales quadratically in sequence length. LongLoRA reduces that cost by changing the attention pattern during training for context extension [Chen et al., 2023]. Our objective was different: the base model already supported the required sequence length, and we wanted the full causal evidence to remain visible during training. We therefore used a training-only blocked computation of the same intended causal attention, while explicitly preserving the qualification that our equivalence tests were not strong enough to prove production-length numerical equivalence.

MLX is designed around Apple Silicon’s unified memory architecture, where CPU and GPU operate on a shared memory pool, and MLX-LM supports LoRA/QLoRA fine-tuning [Hannun et al., 2023, MLX Contributors, 2026]. This makes a 128 GB M4 Max an unusual but useful laboratory: much more memory than a typical laptop GPU, but still constrained enough that long-context activation materialization becomes a first-order systems issue.

4 Behavioral Compilation: A Three-Plane Architecture

We use *behavioral compilation* to describe a specific and limited transformation:

Behavioral compilation is the governed transformation of externally specified task behavior—rubrics, examples, refusal boundaries, decision patterns, and evidence-use conventions—into a parameter update that biases a model toward those behaviors, while leaving mutable authority, source truth, access control, and audit evidence outside the model weights.

The metaphor is intentional. Compilation does not make source code disappear as an authority; it creates an executable artifact from a controlled source. The executable is tested against the specification. When the source changes, one can recompile, roll back, or compare versions. Similarly, an adapter should be treated as a derived behavioral artifact, not the master record of policy.

4.1 The evidence plane

The evidence plane contains what must remain directly addressable: current facts, policy text, source versions, provenance, effective dates, permissions, revocable content, and evidence required for audit. In our repository, benchmark cases are constructed so expected answers do not enter generation; source snapshots and index payloads are hash-bound; and current facts and ACL-sensitive information are explicitly excluded from being delegated to model weights.

This plane is where change should be cheap and semantically explicit. Deleting or superseding a policy should not require proving that every latent trace has disappeared from a checkpoint before the authoritative system can be updated.

4.2 The behavioral weights plane

The behavioral plane contains learned priors that are valuable precisely because they recur: how to decompose a review into obligations, how to distinguish missing work from missing evidence, when to refuse, how to select a disposition, and how to express a bounded answer. This is closer to procedural competence than to a fact database.

The distinction is not absolute. Weights inevitably absorb facts present in training examples. The proposal is normative and architectural: *do not rely on those latent facts as authority*. Train behavior so that the model uses the evidence plane better; do not make the evidence plane unnecessary unless a separate retrieval-free experiment proves that claim.

4.3 The control plane

The control plane owns what should not be probabilistic: schema constraints, source-ID membership, provenance checks, generation budgets, tool permissions, model/revision identity, artifact hashing, and promotion rules. Structured generation is part of this plane. A decoder can enforce that the model returns one valid review object; it cannot determine whether the review is true.

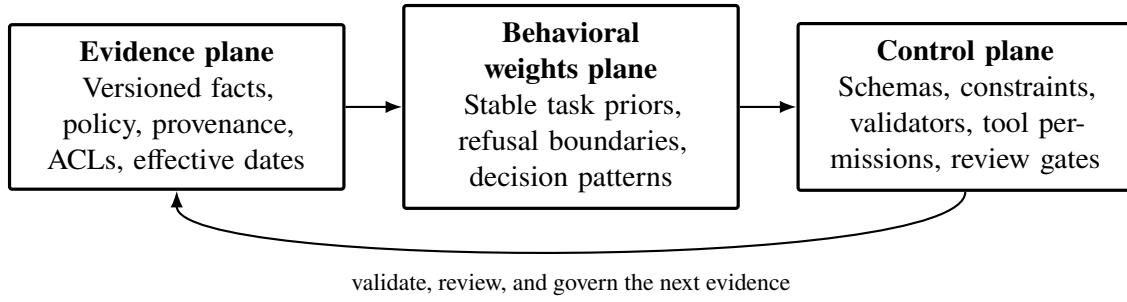


Figure 1: The proposed three-plane architecture. The weights are a derived behavioral artifact. Authority remains external; deterministic control remains outside generation.

This architecture synthesizes an industrial tension visible across Shopify’s reports. One publication argues that the harness is the durable advantage [Deveau, 2026]; another argues that mature production lessons should eventually be translated into parameter space [McNamara and Mazza-Anthony, 2026]. Behavioral compilation treats those statements as complementary rather than contradictory.

5 The Specialization Validity Stack

Our experiments repeatedly failed because we compressed several notions of “correct” into one score. We therefore define a layered validity vector

$$\mathbf{V} = (P, F, D, E, A, G, C), \quad (1)$$

where P is provenance/execution integrity, F structural format validity, D decision correctness, E evidence support, A authority and action safety, G generalization/counterfactual sensitivity, and C continual stability under later updates.

A scalar benchmark score is some projection $s(\mathbf{V})$. The projection may be useful, but it must not erase which layer failed.

L6 Continual stability — Does the behavior survive later updates without interference?
L5 Sensitivity — Does the decision change when material evidence changes?
L4 Authority — Are actions and referrals allowed and safe?
L3 Evidence — Do cited sources support the claims?
L2 Decision — Are labels and dispositions correct?
L1 Structure — Is the output parseable and contract-valid?
L0 Provenance — What code, model, data and artifact actually ran?

Figure 2: The Specialization Validity Stack. Higher layers should not be inferred from lower-layer success. Our structured experiment reached L2 on the exact examples but did not establish L3–L6.

This stack is the methodological core of the paper. The unconstrained adapted model failed L1 while containing a body that matched L2. The constrained decoder made both arms pass L1. The adapter then improved L2 relative to the base. The content review subsequently found L3/L4 defects. No L5 or L6 experiment was run. The apparent contradiction in the results disappears once each claim is attached to the correct layer.

6 Research Trajectory: From Parametric Memory to Task Specialization

6.1 Stage A: closed-book acquisition smoke

The project initially tested whether governed company records could be internalized parametrically. A revised synthetic smoke used the same Qwen3-4B Instruct family, rank-16 LoRA across all 36 layers and seven attention/MLP projections, with 30 optimizer updates. The blinded model-review advisory remained non-promotable and human-review pending, but it provided a useful directional comparison [Dagher, 2026b].

Table 2: Advisory acquisition smoke results (32 acquisition questions per arm). These were model-review labels, not human-certified accuracy.

Arm	Mean score	Fully correct	Hard fails
Base	0.078	2/32	28
Parametric LoRA	0.172	3/32	23
Experimental BM25	0.312	9/32	21
Full context	0.734	22/32	7
Oracle context	0.797	23/32	4

The parametric adapter beat the base on six paired questions, lost on two, and tied on 24, but it failed the pre-registered continuation thresholds. Full context remained the operational winner for the small corpus. The correct interpretation was not that parametric memory had been disproved. It was that this candidate had not justified another cycle, while direct evidence remained far stronger.

That result aligned with the emerging literature: storage in parameters can be feasible, but addressing, continual accessibility, and interference remain hard [O’Neill, 2026, Pulici et al., 2026]. It also suggested a more practical question for local research: instead of trying to replace the evidence channel, can weights learn *how to act on evidence*?

6.2 Stage B: specialize the task before training the student

The next phase focused on a source-grounded deliverable-review task. A larger Qwen model acted as teacher/repairer in development experiments, and advisory evaluators exposed a critical problem: the evaluator itself was not trustworthy. Natural-language obligations were being failed by substring logic; alternative remedies were represented as independent mandatory actions; and unsupported operational consequences could slip through even when citations looked plausible. A blinded advisory over the stopped pilot found both false failures and false acceptances in the deterministic evaluator, leading us to correct task representation rather than train through the noise.

This stage mirrors Shopify’s insistence that quality definition precedes optimization [McNamara and Mazza-Anthony, 2026, McNamara, 2025]. If the rubric does not distinguish the states that matter, more training simply optimizes the wrong contract.

After several synthetic representation experiments, the project stopped prompt iteration. The next useful input was not another prompt variant or a larger teacher. It was a small set of approved development-reference examples that could support a bounded learning-mechanics test.

7 Experimental Design

7.1 Research question

The measured experiment asked:

Can an existing four-bit Qwen3-4B model, adapted with a fixed rank-16 QLoRA configuration on five approved evidence-assisted review examples, learn the approved requirement labels

and case dispositions more completely than the identical unadapted base when both are evaluated through the same structural decoder?

This is an *exact-example fitting* question. It is not a held-out generalization question. The five examples are both the training set and the fitting evaluation set. That limitation is central rather than incidental.

7.2 Frozen configuration

The model was `mlx-community/Qwen3-4B-Instruct-2507-4bit` at pinned revision `50d427756c6b1b2fe0c0a10f67fbda1fc8e82c1b`. Qwen3 supports both thinking and non-thinking modes at the model family level [Yang et al., 2025]; the experiment intended a non-thinking review contract.

Table 3: Measured learning-mechanics configuration.

Base model	Qwen3-4B Instruct, four-bit MLX conversion
Adapter	QLoRA rank 16, scale 2.0, dropout 0.05
Coverage	All 36 layers; q/k/v/o and gate/up/down projections
Optimizer	AdamW, constant 5×10^{-5} learning rate, weight decay 0.01
Batching	Batch size 1; gradient accumulation 1
Random seed	42 before adapter initialization and order
Dataset	5 approved examples per epoch
Epochs	8
Measured updates	Exactly 40
Longest complete row	35,406 tokens
Prompt masking	Prompt excluded from token loss; answer continuation supervised
Hardware	One M4 Max, 40-core integrated GPU, 128 GB unified memory

The longest row contained 33,494 prompt tokens and 1,912 supervised continuation tokens. No example was shortened to make the run fit.

7.3 Target distribution

Across the five approved reviews there were 25 requirement-level labels. Twenty-two were `met`; two were `insufficient_evidence`; one was `not_met`. Three case dispositions were `proceed`; one was `needs_information`; one was `refer_to_source`. This imbalance made the rare negative states disproportionately informative: a majority-class strategy could look competent while missing every boundary condition.

7.4 Why five examples?

Five examples are far too few for a serious generalization estimate, but they were sufficient for the stated mechanics question. The experiment was deliberately bounded after earlier work showed that evaluator defects, data representation, and runtime feasibility could consume more research effort than the optimizer. The aim was to answer one question cleanly before scaling data volume.

This choice is consistent with the spirit, but not the scale, of work showing that high-quality examples can strongly shape behavior [Zhou et al., 2023]; it is also consistent with Shopify’s observation that small, carefully labeled seed sets can drive larger curation systems [Xie, 2026]. Neither analogy turns five examples into a production dataset.

8 Making 35k-Token QLoRA Run Locally

8.1 The stock failure

The stock long-context path failed during the backward pass on the longest example with a Metal insufficient-memory error. The issue was not simply “the four-billion-parameter model is too large.” The dominant materializations were activations.

For the 35,405 shifted input positions, a full attention-score tensor at 32 query heads and float16 scale is approximately

$$32 \times 35,405 \times 35,405 \times 2 \text{ bytes} \approx 75 \text{ GiB.} \quad (2)$$

A full-vocabulary projection over 151,936 vocabulary entries is approximately

$$35,405 \times 151,936 \times 2 \text{ bytes} \approx 10 \text{ GiB.} \quad (3)$$

These are tensor-size estimates, not measurements of total physical memory use. They nonetheless explain why a quantized base can encounter extreme activation pressure.

8.2 Query-blocked causal attention

The training-only runtime processed queries in blocks of 256 while retaining all causally permitted keys and values. A query at absolute position i still attended to all $j \leq i$; this was not a sliding window. The block score tensor was therefore on the order of

$$32 \times 256 \times 35,405 \times 2 \text{ bytes} \approx 0.54 \text{ GiB.} \quad (4)$$

Each block was checkpointed so the backward pass could recompute scores instead of retaining all block matrices simultaneously.

8.3 Project only the supervised positions

The full sequence still traversed the transformer. The optimization occurred at the output projection. If prompt length is P and total row length is L , hidden state i predicts token $i + 1$, so the supervised hidden span is

$$[P - 1, L - 1). \quad (5)$$

For the longest row, $P = 33,494$ and $L = 35,406$, producing 1,912 supervised targets. Only those hidden states were projected to the vocabulary. Projection and cross-entropy were processed in chunks of 256, with per-token losses accumulated in float32 before normalizing by the total supervised-token count.

8.4 Checkpoint once per layer class

A separate retry defect revealed that MLX-LM’s checkpointing helper wraps the layer class’s `__call__`. Applying it once per layer instance repeatedly nested the same class-level wrapper. The corrected installer wraps each relevant class once and refuses a duplicate wrap. This is a small systems detail with a large memory consequence.

8.5 What this implementation does and does not establish

The modified path completed the disposable preflight and then the 40-update measured run from a freshly reloaded base. The final adapter was saved and reload-checked. However, later technical review found material weaknesses in the claimed equivalence validation: the Q/K/V comparison did not explicitly request gradients for all three arguments; the full-model numerical test did not cross the production 256-token chunk boundaries; absolute tolerances were too loose to exclude a missing update at the learning-rate scale; historical executed-code identity was incomplete; and compliance with the intended 96 GiB application-memory ceiling was not established [Dagher, 2026a].

Accordingly, this paper makes a narrower claim:

The training-only runtime completed the recorded long-context QLoRA workload without truncating the five examples or changing the frozen learning settings. It is **not** claimed here to be a formally or numerically proven equivalent implementation of the stock path at production lengths, nor is a 96 GiB whole-process memory cap claimed as established.

9 Evaluation Design: Hold the Interface Constant

9.1 The original unconstrained result

After training, an evaluation-only path reloaded the final adapter and generated one answer for each frozen input. All five generations completed without truncation. Every answer nevertheless began with unexpected XML-like `</tool_call>` markers. Because the official contract required the entire output to be one JSON object, the strict result remained 0/5 complete valid outputs, 0/25 credited labels, and 0/5 dispositions.

A forensic audit found one complete review object after the prefix in every case. Excluding only that prefix, without repairing the JSON or changing any labels, produced a separate post-hoc diagnostic of 25/25 labels and 5/5 dispositions. The diagnostic was explicitly marked `not_original_contract_success`. It could answer a debugging question—“did the body resemble the learned target?”—but it could not replace the original score.

The same audit reconstructed the training boundary and found that the chat template had inserted an empty thinking wrapper into the supervised continuation, while the generation prompt did not contain that wrapper. The generated tool-call markers were different again and were absent from the training message blobs. The exact cause of the markers was not established.

9.2 Structured generation as a controlled intervention

Rather than continue to debug prefixes indefinitely, we introduced a new, separately named structured-inference experiment. The installed `llguidance` library was used as an MLX-LM logits processor. The grammar constrained the declared review schema but did not encode the approved status or disposition for any requirement. Case identifiers and requirement identifiers were restricted only to values already present in the input.

This intervention changed the inference procedure, so it could not be used to retroactively “fix” the official unconstrained result. Instead, it created a matched two-arm experiment:

$$A = \text{base Qwen3-4B} + \text{structured decoder}, \quad (6)$$

$$B = \text{same base} + \text{saved adapter} + \text{same structured decoder}. \quad (7)$$

The decoder’s contribution is the change shared by A and B . The adapter’s measured contribution is the paired difference between B and A under the same structural boundary:

$$\Delta_{\text{weights}} = M(B) - M(A). \quad (8)$$

This design follows the same attribution discipline that production systems need when representation or harness changes are large: do not credit the weights for what the decoder or transpiler provided.

10 Results

10.1 Structured matched comparison

There were ten measured constrained generations: five base and five adapted. All ten began with a JSON object, finished with `stop`, were not truncated, and passed the strict structural contract [Dagher, 2026c].

Table 4: Matched structured-decoding result on the five exact training examples.

Arm	Valid outputs	Labels	Dispositions
Base + structured decoder	5/5	22/25	3/5
Adapter + structured decoder	5/5	25/25	5/5
Adapter increment	0	+3	+2

The three recovered labels were exactly the rare boundary states: the two `insufficient_evidence` requirements and the one `not_met` requirement. The adapted arm also recovered the two non-proceed case dispositions while preserving all 22 majority met labels and all three proceed dispositions.

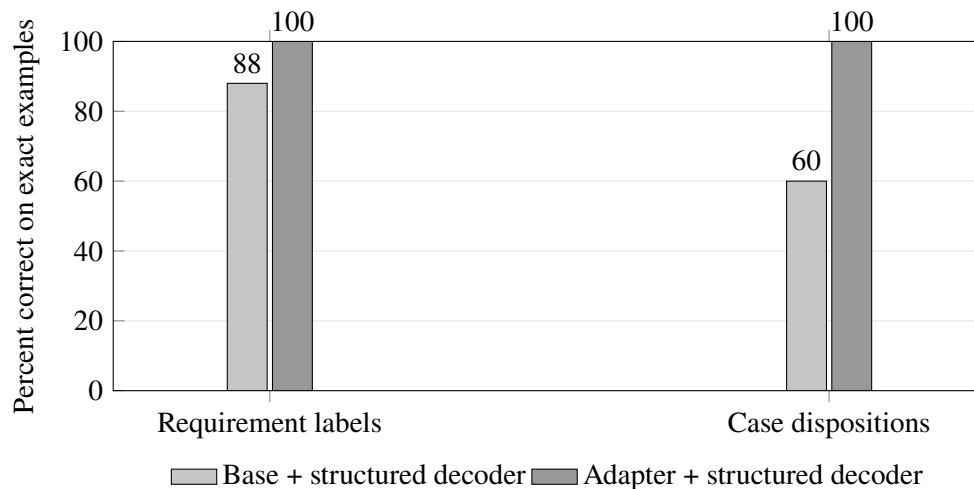


Figure 3: The decoder made both arms structurally valid; the remaining difference is the measured weight effect. Percentages are descriptive only: there are five correlated cases, not an inferential sample.

The experiment therefore supports *exact-example decision fitting*: under a common structured interface, the saved adapter changed the model’s decisions toward all approved labels and dispositions on the examples it had seen. It does not establish that the model learned a general review policy.

10.2 Why the 0/5 and the 25/25 are both valid results

The historical 0/5 and the later 25/25 answer different questions. The 0/5 measures compliance with the original whole-response interface. The 25/25 measures requirement labels inside valid structured reviews under a new matched decoder. The body-only 25/25 measures what was present inside the failed original responses after a post-hoc structural exclusion. None should overwrite another.

This distinction is not bookkeeping trivia. It demonstrates why the validity stack is necessary. A system can fail format, fit decision labels, and still produce unsupported reasoning.

11 Semantic Review: Perfect Decisions Were Not Perfect Reviews

After the structured comparison, the ten measured answers were packaged for a blinded model-advisory review. Arm identity, adapter status, previous scores, timing, and approved labels were hidden. The reviewer assessed evidence support, omitted or invented requirements, the distinction between missing work and missing evidence, authority/referral handling, next-action safety, and internal consistency.

The review was not human attestation and was not fully independent: the reviewing system had participated in earlier design discussion and had seen aggregate project summaries. In addition, ten governed source artifacts were initially unavailable as exact frozen bytes. The review contract therefore

required overall numerical scores to remain null whenever a listed source was withheld. All ten overall scores remained null; this was not equivalent to ten failures.

Even with that limitation, contradictions visible in available evidence could still be recorded. Across the ten answers, the advisory found at least one unsupported observation or attribution in every answer. Seven correctly distinguished missing work from missing evidence and three did not. Authority/referral handling was assessed as two correct, six incorrect, and two unclear. Next actions were four supported, one unsafe-or-contradictory, and five unclear. Internal consistency was six agreements and four disagreements.

After those findings were frozen, the maintainer joined them to the private arm mapping locally. The paired findings were more informative than a pooled count.

Table 5: Paired qualitative findings after local unblinding. “Improvement” and “regression” are descriptive model-advisory observations, not certified semantic scores.

Case	Base finding	Adapted finding
DEV-001	Overstated machine authority and evidence path; invented stress/production gaps; unclear action	More cautious evidence request and supported action, but inaccurate parser claims and unsupported owner-confirmation language
DEV-002	Misstated an availability mock and overclaimed production invariance; unclear action	Fixture/integrity account largely supported; still invented owner-confirmed coverage; supported action
DEV-003	Reversed the required Phase-B access rule; next action contradicted the workflow	Two-phase conclusion mostly supported; false manifest-hash attribution; supported action
DEV-004	Overstated runtime/identity evidence; impossible line citation; incorrect precision claim	Correctly noticed missing execution evidence, but invented interim authority and confused the requirement boundary
DEV-005	Substantial implementation support, but wrong section citations and ungrounded production check	Mostly sound implementation citations, but unsupported owner-confirmed exemption language

The most consequential paired result was DEV-003. The base model recommended hiding historical results during the phase that was defined to compare against those historical results. The advisory marked that action unsafe-or-contradictory in the workflow sense. The adapted answer removed that operational contradiction and reached a more defensible conclusion, but still cited the wrong artifact for a hash binding. This is exactly the type of result a scalar label score hides: a real improvement and a real remaining defect can coexist in the same answer.

DEV-004 showed the reverse risk. The adapted answer correctly recognized missing runtime evidence and selected the approved referral disposition, yet also asserted interim authority that the supplied policy did not grant. The final decision label was right; part of the justification was not.

The semantic review therefore changes the interpretation without erasing the structured result:

The adapter learned the approved *decision boundary* on the five examples more completely than the base under an identical structured decoder. It did not thereby establish reliable source-grounded reasoning, correct authority semantics, or safe operational recommendations.

12 What We Reproduced—and What We Did Not

Calling this work a “reproduction of Shopify” would be inaccurate. The compute, data, model size, product surface, and online feedback loop are fundamentally different. What we reproduced was a set of methodological constraints at laboratory scale.

12.1 What transferred

Harness before weights. We spent substantial effort on split integrity, evaluator semantics, output contracts, and provenance before allowing the bounded training run. Shopify similarly emphasizes a durable harness and quality contract [Deveau, 2026, McNamara and Mazza-Anthony, 2026].

Representation and parity matter. Our training-serving token audit and structured-decoding experiment confirmed that formatting can dominate the measured outcome independently of learned decisions, echoing Shopify Flow’s representation and parity findings [Chaiwachirasak and colleagues, 2026].

Hard negatives matter disproportionately. The adapter’s measurable gain was entirely on rare missing-evidence/unmet/non-proceed states, consistent with Shopify’s emphasis on refusal data and failure mining [Xie, 2026].

Procedural validation and semantic judgment are different. The grammar guaranteed valid JSON; it did not guarantee truth. Shopify similarly separates programmatic validation from semantic judges and has reported reward hacking when the reward contract is incomplete [McNamara, 2025].

Smaller specialized models can be interesting for reasons other than benchmark prestige. Shopify has repeatedly reported production benefits from smaller specialized open models in Flow and catalogue systems [Chaiwachirasak and colleagues, 2026, Guindon, 2025]. Our local experiment explores the same direction at a much smaller and non-production scale.

12.2 What did not transfer

We had no millions of production conversations, no user activation metric, no human-calibrated production judge, no online A/B test, no continual training loop, no GRPO phase, no gisting deployment, and no held-out real-work suite. Shopify’s strongest claims depend precisely on those systems. Our experiment cannot inherit them by analogy.

The workstation is therefore not the product. It is the microscope. Its value is that ten generations, five examples, one pinned model, one adapter, one decoder, and one sequence of approvals can be inspected in enough detail to attribute failure modes. Production scale answers a different question: whether the loop continues to work under evolving traffic, economics, and organizational constraints.

13 Implications for Enterprise AI Architecture

13.1 Do not use weights as the authoritative record

The strongest architectural conclusion from the combined literature and our own experiments is not “never put facts in weights.” The literature demonstrates that facts can be edited or internalized [Meng et al., 2022a,b, Kou et al., 2026]. The conclusion is that *authority and addressability are separate requirements from storage*. O’Neill’s continual-write results show that a fact can remain latent yet become behaviorally inaccessible [O’Neill, 2026]. Pulici et al. show that stored adapter knowledge can exist without a reliable selector [Pulici et al., 2026]. In an enterprise system, a fact that cannot be deterministically located, superseded, permissioned, or audited is a poor sole source of truth even if the model can sometimes answer with it.

13.2 Compile stable behavior, retrieve mutable evidence

A more promising division of labor is to place stable behavioral invariants in the adapter: how to ask for missing evidence, when to refuse, what constitutes a valid referral, how to decompose a task, and which response structure the organization expects. The live evidence still arrives from a source system.

This is not merely a governance concession. It may also be a better learning problem. Behavior can recur across many documents and workflows, giving adaptation repeated signal. A current policy value may change tomorrow and is intrinsically harder to treat as a durable parametric target.

13.3 Structured generation is an interface, not an epistemology

Our most visually impressive result—0/5 to 5/5 valid outputs—came entirely from a decoder intervention that helped both arms. It would have been easy to describe this as a dramatic fine-tuning success if the base control had been omitted. Structured decoding should be understood as an interface/type system. It eliminates malformed states and narrows the action space. It does not supply the missing evidence or make an unsupported statement true.

13.4 The next scarce resource is validated feedback

Once basic training becomes accessible, the bottleneck moves to data quality and evaluation quality. Shopify’s production work repeatedly reaches the same conclusion: high-quality seeds, calibrated judges, hard negatives, and continuous gap discovery are more durable assets than a one-time training recipe [Xie, 2026, McNamara and Mazza-Anthony, 2026]. Our project reached that bottleneck almost immediately. Five examples were enough to prove learning mechanics; they were nowhere near enough to prove a robust skill.

14 Threats to Validity and Non-Claims

14.1 Training-set evaluation

The headline 25/25 and 5/5 result is measured on the five examples used for training. It demonstrates fit, not generalization. The 25 requirement labels are nested within five cases and cannot be treated as 25 independent trials. No statistical significance claim is made.

14.2 Model-advisory semantics

The content review was performed by a model advisory, not independent human reviewers. The reviewer had prior project context, though arm identities were hidden. Overall scores were null because five frozen patch/diff artifacts remained unrecovered under the review contract. Dimension-level defects that contradicted available evidence remain informative, but they are not a certified pass/fail rate.

14.3 Long-context runtime validation

The custom training path is subject to the public qualification described earlier. In particular, the Q/K/V gradient check, chunk-boundary coverage, tolerance sensitivity, historical code identity, and memory-cap evidence are incomplete. The paper therefore avoids the word *exact* as a proven numerical property of the full production-length runtime, even though the original helper name and technical note used that label.

14.4 Provenance gaps

A later closure review found that an execution field named `freeze_manifest_sha256` actually contained the measurement-contract digest, and complete per-run freeze snapshots had not been copied into the run directories. The current files and recorded hashes permit partial binding, not reconstruction of every historical byte. We preserve this as a provenance limitation rather than silently repairing history.

14.5 Private data and reproducibility

The public repository excludes model weights, raw private evidence, prompts, targets, token IDs, owner/approval metadata, private mappings, and blinded review packets. Public code and sanitized reports allow inspection of the method but do not enable independent reproduction of every private-case judgment.

14.6 Explicit non-claims

This paper does not claim:

- deployment readiness or production safety;
- closed-book company knowledge acquisition;
- generalization to unseen work;
- a replacement for retrieval, tools, databases, provenance, or human sign-off;
- reliable continual learning or deletion from weights;
- a certified semantic judge;
- equivalence to Shopify’s scale, quality, economics, or production feedback loop;
- numerical equivalence of the modified long-context runtime at the measured production sequence lengths.

15 Research Agenda

The learning-mechanics phase is closed. The next useful work should increase epistemic value rather than add more optimizer cycles.

15.1 Counterfactual evidence sensitivity

The immediate next experiment should hold the task and decoder fixed while changing the evidence. Material changes should cause the appropriate decision to change; non-material paraphrases or reorderings should leave it stable. This is the first direct test of whether the adapter learned an evidence-conditioned rule rather than simply a familiar output pattern.

A useful design would create paired transformations $x \rightarrow x'$ with a pre-registered expected relation:

$$D(x') = \begin{cases} D(x), & \text{if the transformation is semantically irrelevant,} \\ D^*(x'), & \text{if the transformation changes a governing condition.} \end{cases} \quad (9)$$

This would move the experiment from L2 decision fitting toward L5 sensitivity in the validity stack.

15.2 Approved held-out real work

Synthetic cases were useful for isolating representation defects, but a serious specialization claim requires real workflow examples with owner-reviewed evidence, frozen before generation. The evaluation should separate development cases from a held-out final set and preserve ambiguity rather than forcing a label when policy itself is underspecified.

15.3 Hard-negative curation

The observed gain concentrated on rare negative states. Future data should therefore not be dominated by successful *met/proceed* trajectories. A curation process should deliberately seek ambiguity, missing evidence, missing capability, authority conflicts, and invalid next actions, taking inspiration from Shopify’s refusal pipeline while preserving local human governance [Xie, 2026].

15.4 Calibrated semantic evaluation

Before using an LLM judge as a reward signal, its rubric should be calibrated against multiple human reviewers and inter-annotator agreement, following the principle demonstrated in Sidekick’s evaluation work [McNamara, 2025, McNamara and Mazza-Anthony, 2026]. Deterministic checks should remain responsible for typed structure and provenance; semantic judges should not be asked to re-decide what code can establish exactly.

15.5 SFT before policy optimization

Shopify’s continual loop uses repaired trajectories for supervised fine-tuning before reinforcement-style optimization [McNamara and Mazza-Anthony, 2026]. Our local project should follow the same ordering only after the rubric is trustworthy. GRPO or similar methods are inappropriate while the reward can still confuse correct labels with unsupported rationale.

15.6 Gisting only after behavior is stable

Prompt compression is attractive because long system instructions are a recurring inference tax. Gist-token research and Shopify’s production deployment show that stable context can be compressed substantially [Mu et al., 2023, Mazza-Anthony and Vegna, 2026]. But compression should follow behavioral validation, not precede it. Otherwise a shorter prompt can make an already-ambiguous policy harder to diagnose.

15.7 Continual updates and interference

Only after held-out competence exists should the project study sequential specialization. That experiment must measure retention, not merely the newest task. O’Neill’s results and LoRA spectral analyses provide strong reasons to expect interference under sequential writes [O’Neill, 2026, Shuttleworth et al., 2024].

16 Conclusion

The motivating industrial idea was ambitious: mature agent systems should eventually turn repeated production lessons into model parameters instead of letting every lesson accumulate forever in prompts and control code. We asked whether a meaningful fragment of that process could be studied on a single local machine without pretending that the machine, data, or results were production scale.

The answer is yes—but the useful result is more nuanced than a fine-tuning success story.

A closed-book parametric acquisition candidate improved slightly over its base but remained far behind direct context, pushing the project toward task specialization rather than factual replacement. A five-example QLoRA run then fit a narrow deliverable-review task on a single M4 Max despite 35k-token training rows. The original unconstrained evaluation failed its output contract. Structured decoding then made both base and adapter structurally valid and isolated a genuine adapter increment: three additional requirement labels and two additional case dispositions on the exact examples. Semantic review subsequently showed that some answers still contained unsupported source claims, authority errors, and an operationally contradictory recommendation.

These are not contradictory outcomes. They occupy different layers of validity.

The result suggests a design direction for enterprise AI: **keep authority addressable, compile behavior selectively, and evaluate the compiled artifact as a system rather than as a single accuracy number**. We call that direction behavioral compilation.

The distinctive asset is therefore not the adapter checkpoint. It is the governed loop that produces and challenges it:

evidence → approved behavior → weights → constrained execution → layered review → new evidence.

On a workstation, that loop can be slow. It can also be unusually legible. Every failure in this study—memory exhaustion, evaluator false failures, format violation, provenance ambiguity, unsupported semantic claims—made the final architecture more conservative and more precise.

The workstation is not the product. It is the microscope.

Reproducibility and Artifact Availability

The public implementation is available in the `senoni-research/enterprise-memory-mlx` repository. The principal public milestone discussed here is `commit`

`419c4c2069c6760830376af6dd5b92179355c5dd`.

The repository contains the long-context training implementation, structured-inference implementation, model-free tests, a sanitized structured-inference result report, and the qualification to the long-context technical note. Public CI passed on the milestone branch. Generated private artifacts, model weights, raw review evidence, targets, token IDs, approval metadata, and unblinding maps are intentionally excluded.

Acknowledgment of Source Types

This paper draws on both academic literature and industrial engineering reports. Shopify Engineering sources are cited as first-party descriptions of Shopify’s production systems and observed engineering results; they are not presented as peer-reviewed scientific studies. Several 2026 research sources cited here are recent arXiv preprints and should likewise be read as pre-publication research.

References

- Sahil Bhatia, Jie Qiu, Niranjan Hasabnis, Sanjit A. Seshia, and Alvin Cheung. Verified code transpilation with llms. In *Advances in Neural Information Processing Systems*, 2024. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/48bb60a0c0aebb4142bf314bd1a5c6a0-Abstract-Conference.html.
- Ted Chaiwachirasak and colleagues. Flow generation through natural language: An agentic modeling approach. Shopify Engineering, April 2026. URL <https://shopify.engineering/fine-tuning-agent-shopify-flow>. Updated 22 April 2026.
- Yukang Chen, Shengju Qian, Haotian Tang, Xin Lai, Zhijian Liu, Song Han, and Jiaya Jia. Longlora: Efficient fine-tuning of long-context large language models. *arXiv preprint arXiv:2309.12307*, 2023. URL <https://arxiv.org/abs/2309.12307>.
- Philippe Dagher. Qualification to the exact-memory technical note. Enterprise Memory MLX public research note, 2026a. URL <https://github.com/senoni-research/enterprise-memory-mlx/blob/419c4c2069c6760830376af6dd5b92179355c5dd/docs/exact-memory-long-context-qlora.qualification-2026-09-11.md>.
- Philippe Dagher. Advisory smoke result – 1 september 2026. Enterprise Memory MLX public research note, 2026b. URL <https://github.com/senoni-research/enterprise-memory-mlx/blob/419c4c2069c6760830376af6dd5b92179355c5dd/docs/results/2026-09-01-smoke-advisory.md>.
- Philippe Dagher. Qwen4b structured exact-example fitting. Enterprise Memory MLX public research note, 2026c. URL <https://github.com/senoni-research/enterprise-memory-mlx/blob/419c4c2069c6760830376af6dd5b92179355c5dd/docs/qwen4b-structured-inference-v1.report.md>.
- Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. Qlora: Efficient finetuning of quantized llms. *Advances in Neural Information Processing Systems*, 2023. URL <https://arxiv.org/abs/2305.14314>.

- Zack Deveau. Building an agentic harness that outlasts the model. Shopify Engineering, July 2026. URL <https://shopify.engineering/building-an-agentic-harness-that-outlasts-the-model>. Published 29 July 2026.
- Shengda Fan, Xin Cong, Yuepeng Fu, Zhong Zhang, Shuyan Zhang, Yuanwei Liu, Yesai Wu, Yankai Lin, Zhiyuan Liu, and Maosong Sun. Workflowlm: Enhancing workflow orchestration capability of large language models. *arXiv preprint arXiv:2411.05451*, 2024. URL <https://arxiv.org/abs/2411.05451>.
- Audrey-Anne Guindon. Leveraging multimodal llms for shopify’s global catalogue: Recap of expo talk at iclr 2025. Shopify Engineering, July 2025. URL <https://shopify.engineering/leveraging-multimodal-llms>. Published 16 July 2025.
- Awni Hannun, Jagrit Digani, Angelos Katharopoulos, and Ronan Collobert. MLx: Efficient and flexible machine learning on apple silicon, 2023. URL <https://github.com/ml-explore/mlx>.
- Cheng-Yu Hsieh, Chun-Liang Li, Chih-Kuan Yeh, Hootan Nakhost, Yasuhisa Fujii, Alexander Ratner, Ranjay Krishna, Chen-Yu Lee, and Tomas Pfister. Distilling step-by-step! outperforming larger language models with less training data and smaller model sizes. In *Findings of the Association for Computational Linguistics: ACL 2023*, 2023. URL <https://arxiv.org/abs/2305.02301>.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models. *arXiv preprint arXiv:2106.09685*, 2021. URL <https://arxiv.org/abs/2106.09685>.
- Qian Kou, Xiaofeng Shi, Xiaosong Qiu, and Hua Zhou. Inject, align, recover: Staged post-training for retrieval-free document knowledge internalization. *arXiv preprint arXiv:2608.20281*, 2026. URL <https://arxiv.org/abs/2608.20281>.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Kuttler, Mike Lewis, Wen tau Yih, Tim Rocktaschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks. In *Advances in Neural Information Processing Systems*, 2020. URL <https://arxiv.org/abs/2005.11401>.
- Cody Mazza-Anthony and Paige Vegna. Gisting: Compressing llm agent context to increase throughput and reduce cost. Shopify Engineering, August 2026. URL <https://shopify.engineering/gisting>. Published 19 August 2026.
- Andrew McNamara. Building production-ready agentic systems: Lessons from shopify sidekick. Shopify Engineering, August 2025. URL <https://shopify.engineering/building-production-ready-agentic-systems>. Based on an ICML 2025 talk with Ben Lafferty and Michael Garner.
- Andrew McNamara and Cody Mazza-Anthony. Sidekick’s continual learning loop. Shopify Engineering, August 2026. URL <https://shopify.engineering/sidekicks-continual-learning-loop>. Published 5 August 2026.
- Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in gpt. *Advances in Neural Information Processing Systems*, 2022a. URL <https://arxiv.org/abs/2202.05262>.
- Kevin Meng, Arnab Sen Sharma, Alex Andonian, Yonatan Belinkov, and David Bau. Mass-editing memory in a transformer. *arXiv preprint arXiv:2210.07229*, 2022b. URL <https://arxiv.org/abs/2210.07229>.
- Eric Mitchell, Charles Lin, Antoine Bosselut, Chelsea Finn, and Christopher D. Manning. Fast model editing at scale. *arXiv preprint arXiv:2110.11309*, 2021. URL <https://arxiv.org/abs/2110.11309>.
- MLX Contributors. MLx-lm: Run and fine-tune large language models on apple silicon, 2026. URL <https://github.com/ml-explore/mlx-lm>.

- Federico Mora, Justin Wong, Haley Lepe, Sahil Bhatia, Karim Elmaaroufi, George Varghese, Joseph E. Gonzalez, Elizabeth Polgreen, and Sanjit A. Seshia. Synthetic programming elicitation for text-to-code in very low-resource programming and formal languages. *arXiv preprint arXiv:2406.03636*, 2024. URL <https://arxiv.org/abs/2406.03636>.
- Jesse Mu, Xiang Lisa Li, and Noah Goodman. Learning to compress prompts with gist tokens. *Advances in Neural Information Processing Systems*, 2023. URL <https://arxiv.org/abs/2304.08467>.
- Charles O’Neill. Can a language model learn facts continually in its weights? *arXiv preprint arXiv:2607.11020*, 2026. URL <https://arxiv.org/abs/2607.11020>.
- Martino M. L. Pulici, Cuong Xuan Chu, Evgeny Kharlamov, and Volker Tresp. A storage-retrieval gap in parametric knowledge graph memory. *arXiv preprint arXiv:2608.25489*, 2026. URL <https://arxiv.org/abs/2608.25489>.
- Reece Shuttleworth, Jacob Andreas, Antonio Torralba, and Pratyusha Sharma. Lora vs full fine-tuning: An illusion of equivalence. *arXiv preprint arXiv:2410.21228*, 2024. URL <https://arxiv.org/abs/2410.21228>.
- Shuang Xie. Teaching sidekick to say no: Automated data curation with llm judge consensus. *Shopify Engineering*, June 2026. URL <https://shopify.engineering/sidekick-curation>. Published 15 June 2026.
- An Yang et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025. URL <https://arxiv.org/abs/2505.09388>.
- Chunting Zhou, Pengfei Liu, Puxin Xu, Srini Iyer, Jiao Sun, Yuning Mao, Xuezhe Ma, Avia Efrat, Ping Yu, Lili Yu, Susan Zhang, Gargi Ghosh, Mike Lewis, Luke Zettlemoyer, and Omer Levy. Lima: Less is more for alignment. *arXiv preprint arXiv:2305.11206*, 2023. URL <https://arxiv.org/abs/2305.11206>.

A Claims Ledger

Table 6: Claims ledger for the reported study.

Claim	Status	Boundary
The modified runtime completed the five-example, 40-update long-context QLoRA workload.	Supported	Recorded local execution; no claim of full numerical equivalence.
The longest complete row was 35,406 tokens.	Supported	Loader counts and frozen settings; no truncation in the measured run.
The original unconstrained adapted evaluation achieved 0/5 complete valid outputs.	Supported	Strict whole-text JSON contract; historical result preserved.
The body-only diagnostic matched 25/25 labels and 5/5 dispositions.	Supported diagnostic	Post-hoc structural exclusion; explicitly not original-contract success.
Under identical structured decoding, the base achieved 22/25 labels and 3/5 dispositions.	Supported	Five exact training examples; descriptive, not inferential.
Under identical structured decoding, the adapter achieved 25/25 labels and 5/5 dispositions.	Supported	Five exact training examples; no generalization claim.
The adapter therefore improved exact-example decisions relative to the base.	Supported	+3 requirement labels, +2 case dispositions under matched decoder.
The complete adapted reviews are semantically correct.	Not established	Model-advisory review found unsupported claims; overall semantic scores remained null.
The adapter generalizes to unseen real work.	Not tested	Requires frozen held-out cases.
The adapter acquired closed-book company knowledge.	Not established	Evidence remained available in the specialization task; separate acquisition smoke failed continuation rule.
The long-context runtime is numerically equivalent to stock MLX-LM at 35k tokens.	Not established	Validation gaps documented in public qualification.
The training process respected a verified 96 GiB application-memory cap.	Not established	Historical process footprint and swap attribution incomplete.
The system is deployment-ready.	Not claimed	No production, human adjudication, or held-out generalization evidence.

B Experimental Timeline

- Governed acquisition smoke.** Parametric LoRA improved over base but remained far behind full/oracle context and failed continuation thresholds.
- Task-specialization development.** Teacher/repair and evaluator experiments exposed representation and scoring defects; training remained blocked.
- Five-example learning-mechanics authorization.** Exact model, examples, optimizer settings, update budget, and stop conditions were frozen.

4. **Long-context engineering.** Stock backward path failed; training-only query blocking and supervised-position projection enabled the bounded run.
5. **Measured 40-update run.** Fresh base and adapter; eight exposures per case; final adapter saved/reload checked.
6. **Unconstrained evaluation.** All five outputs failed strict whole-text JSON because of unexpected protocol markers; body-only diagnostic recorded separately.
7. **Structured-inference comparison.** Same decoder for base and adapter; ten measured generations; adapter recovered the five decision errors made by the base.
8. **Blinded semantic advisory.** Evidence/authority/action defects recorded without arm identities; overall scores left null under missing-source rule.
9. **Local unblinding and closure.** Paired qualitative comparison preserved; no further generation or training authorized.

C Proposed Minimum Next Experiment

A credible next experiment should be smaller than a production flywheel but stronger than a training-set fitting check. We propose a *counterfactual evidence-sensitivity matrix* for a held-out set of owner-reviewed cases. Each base case receives at least two controlled mutations:

1. a **non-material mutation** (reordering, paraphrase, irrelevant evidence) for which the decision should remain invariant;
2. a **material mutation** (add/remove decisive evidence, resolve/create an authority conflict) for which one or more requirement labels and possibly the disposition should change.

Both unadapted and adapted models should run under the same structured decoder. The evaluator should score the entire validity vector, not just the final label. No additional training should occur until this experiment establishes whether the saved adapter is actually evidence-sensitive.